

Course: B. Tech. Branch : Mechanical & Allied Engineering Semester :VI

Subject Code & Name: Machine Design - II (BTMEC602_Y19)

Max Marks: 60

Date:- 18/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following. 12

- A) A single-row deep groove ball bearing is subjected to a pure radial force of 3 KN from a shaft that rotates at 600 rpm. The expected life L_{10h} of the bearing is 30000 h. The minimum acceptable diameter of the shaft is 40 mm. Select a suitable ball bearing for this application. CO1 6
- B) Write the procedure for selection of bearing from manufacturers catalogue CO1 6
- C) Compare Rolling contact bearing with sliding contact bearing. CO1 6

Q.2 Solve Any Two of the following. 12

- A) A pair of spur gear with 20° full-depth involute teeth consists of a 20 teeth pinion meshing with a 41 teeth gear. The module is 3 mm while the face width is 40 mm. The material for pinion as well as gear is steel with ultimate tensile strength of 600 N/mm². The pinion rotates at 1450 rpm and the service factor for the application is 1.75. Assume that the velocity factor accounts for the dynamic load and factor of safety is 1.5. Determine the rated power that the gears can transmit. The Lewis form factor is 0.32 for 20 teeth. CO2 6
- B) Discuss the different modes of gear tooth failure. CO2 6
- C) Do the force analysis of helical gear with neat sketch CO2 6

Q. 3 Solve Any Two of the following. 12

- A) A pair of bevel gears, with 20° pressure angle, consists of a 20 teeth pinion meshing with a 30 teeth gear. The module is 4.0 mm, while the face width is 40 mm. the material for the pinion and gear is steel 50C4 ($S_{ut} = 750$ N/mm²). The gear teeth are lapped and ground (Class-3) and the surface hardness is 400 BHN. The pinion rotates at 500 rpm and receives 2.5 KW power from the electric motor. The starting torque of the motor is 150% of the rated torque. Determine the factor of safety against bending failure and against pitting. CO3 6
- B) A worm drive transmits 15 kW at 2000 r.p.m. to a machine carriage at 75 r.p.m. The worm is triple threaded and has 65 mm pitch diameter. The worm gear has 90 teeth of 6 mm module. The tooth form is to be 20° full depth involute. The coefficient of friction between the mating teeth may be taken as 0.10. Calculate: 1. tangential force acting on the worm; 2. axial thrust and separating force on worm; and 3. efficiency of the worm drive. CO3 6

C)	Explain the terms, Axial pitch, Lead, Lead angle, Tooth pressure angle, Normal pitch and Helix angle, used in Worm Gearing.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	What is the Condition for the Transmission of Maximum Power in flat belt drives? Derive An expression.	CO4	6
B)	State Design Procedure of Chain Drive.	CO4	6
C)	A single cylinder double acting steam engine develops 150 kW at a mean speed of 80 r.p.m. The coefficient of fluctuation of energy is 0.1 and the fluctuation of speed is $\pm 2\%$ of mean speed. If the mean diameter of the flywheel rim is 2 metres and the hub and spokes provide 5 percent of the rotational inertia of the wheel, find the mass of the flywheel and cross-sectional area of the rim. Assume the density of the flywheel material (which is cast iron) as 7200 kg / m^3 .	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	A rope drum of an elevator having 650 mm diameter is fitted with a brake drum of 1 m diameter. The brake drum is provided with four cast iron brake shoes each subtending an angle of 45° . The mass of the elevator when loaded is 2000 kg and moves with a speed of 2.5 m / s. The brake has a sufficient capacity to stop the elevator in 2.75 metres. Assuming the coefficient of friction between the brake drum and shoes as 0.2, find: 1. width of the shoe, if the allowable pressure on the brake shoe is limited to 0.3 N/mm^2 ; and 2. heat generated in stopping the elevator.	CO5	6
B)	A single plate clutch, effective on both sides, is required to transmit 25 kW at 3000 r.p.m. Determine the outer and inner diameters of frictional surface if the coefficient of friction is 0.255, ratio of diameters is 1.25 and the maximum pressure is not to exceed 0.1 N/mm^2 . Also, determine the axial thrust to be provided by springs. Assume the theory of uniform wear.	CO5	6
C)	What is autofrettage in Cylinders and describe the way to reduce it.	CO6	6

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